

Adversarial Machine Learning Against Network Intrusion Detection

Live Project Screenshots Documentation Project Execution Flow Report

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Step 1: Home Page



Figure 1. Home Page

Description

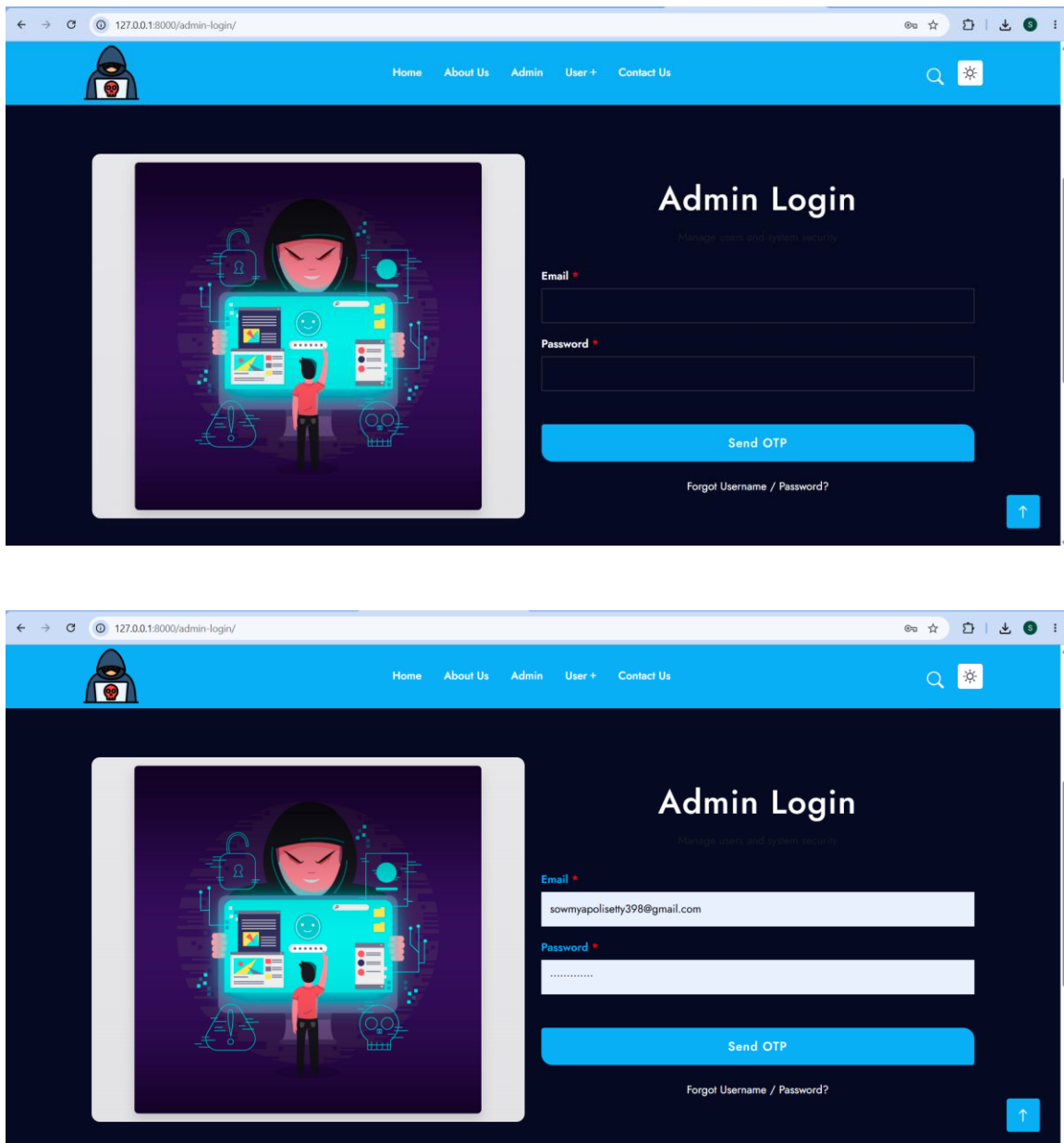
The home page is the entry point of the proposed system.

It provides project information and navigation to Admin and User modules.

Users can access login, registration, and contact options from this page.

It serves as the starting point of the network intrusion detection workflow.

Step 2: Admin Login



The image displays two screenshots of a web application's Admin Login page. The page has a dark blue background with a light blue header. The header contains a navigation menu with links: Home, About Us, Admin, User +, and Contact Us. A search icon and a settings icon are also present in the header. The main content area features a large illustration on the left showing a person in a hoodie interacting with a glowing screen displaying various icons. To the right of the illustration is the 'Admin Login' section. This section includes the title 'Admin Login' and the subtitle 'Manage users and system security'. Below the subtitle are two input fields: 'Email' and 'Password'. A blue button labeled 'Send OTP' is positioned below the password field. At the bottom of the login section is a link that says 'Forgot Username / Password?'. A small blue button with an upward arrow is located at the bottom right of the login section. The top screenshot shows the form fields empty, while the bottom screenshot shows the 'Email' field filled with the text 'sowmyapolisetty398@gmail.com'.

127.0.0.1:8000/admin-login/

Home About Us Admin User + Contact Us

Admin Login

Manage users and system security

Email *

Password *

Send OTP

Forgot Username / Password?

sowmyapolisetty398@gmail.com

Figure 2. Admin Login

Description

The administrator enters the registered email and password to log in. An OTP-based authentication mechanism ensures secure access. Only authorized administrators can access the admin dashboard. This prevents unauthorized access to system management functions..

Step 3: Admin OTP Verification

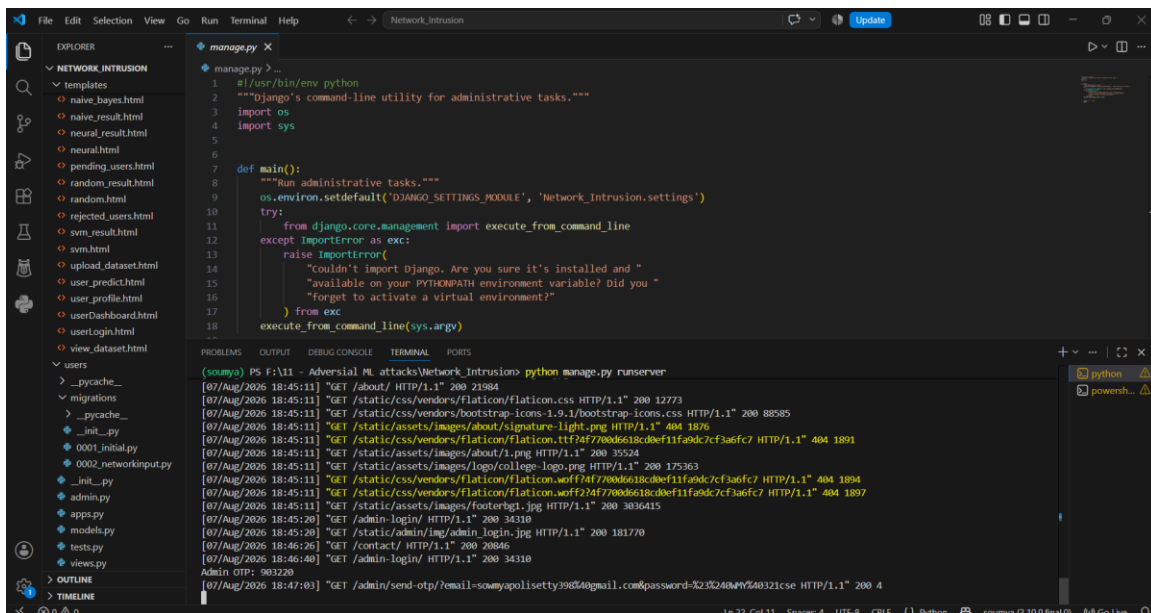
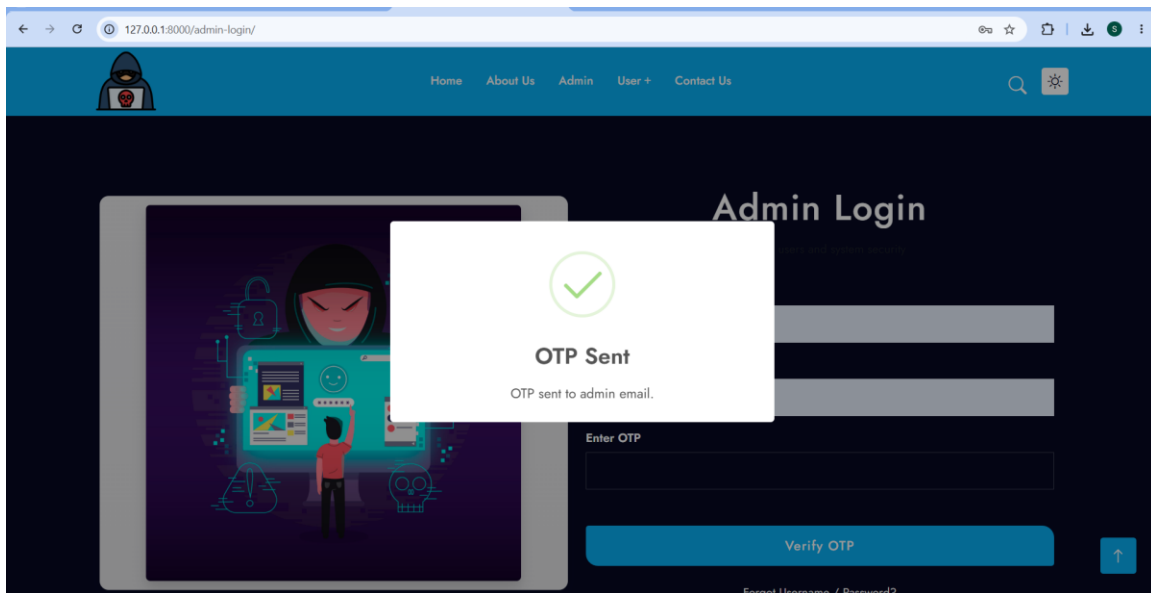


Figure 3. Admin OTP Verification

Description

An OTP is sent to the administrator's registered email address. The administrator enters the OTP to verify their identity. Successful verification grants secure access to the dashboard. This adds an additional layer of authentication.

Step 4: Admin Dashboard

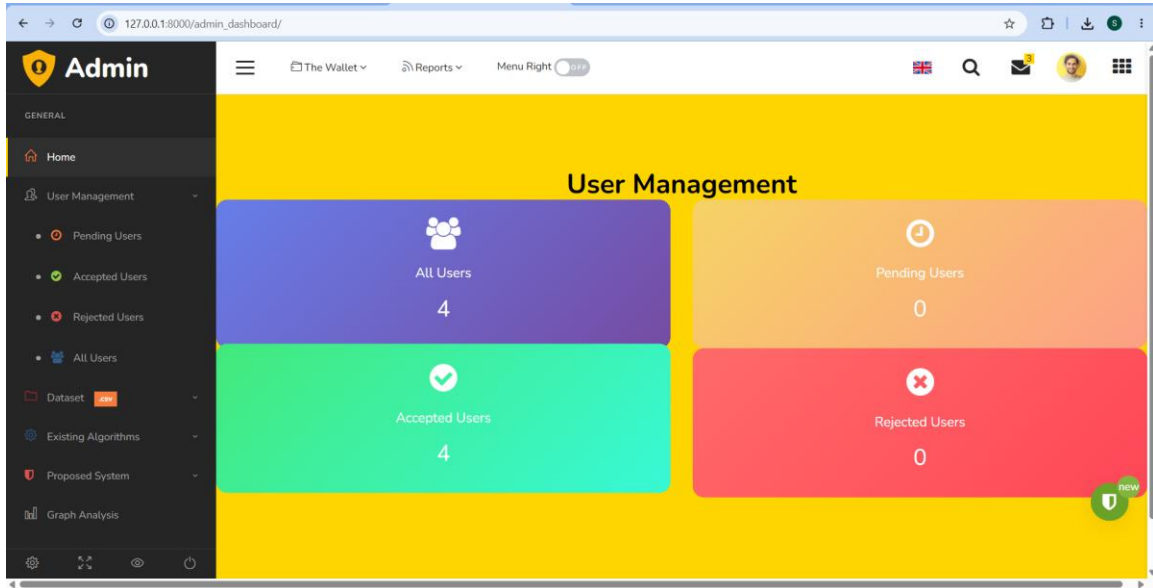


Figure 4. Admin Dashboard

Description

The Admin Dashboard serves as the central control panel of the proposed intrusion detection system.

It provides access to user management, dataset upload, machine learning model execution, and performance analysis modules.

The administrator can manage user accounts, train multiple classification algorithms, and monitor system performance.

Graphical comparisons and evaluation metrics help analyze the effectiveness of each model. This dashboard enables efficient monitoring and administration of the entire application.

Step 5: Pending Users(User Management)

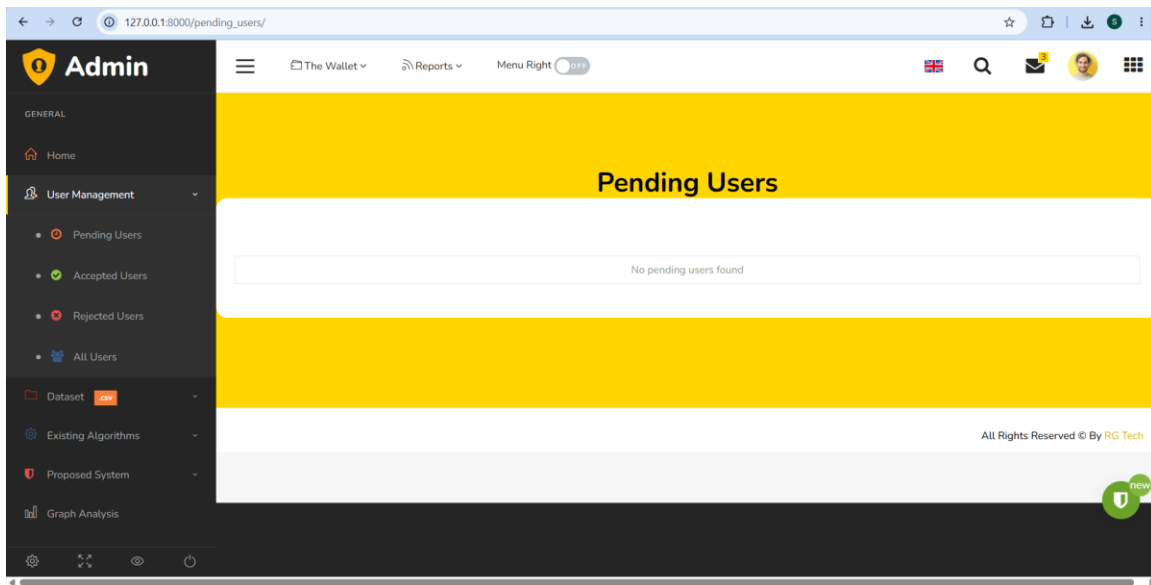
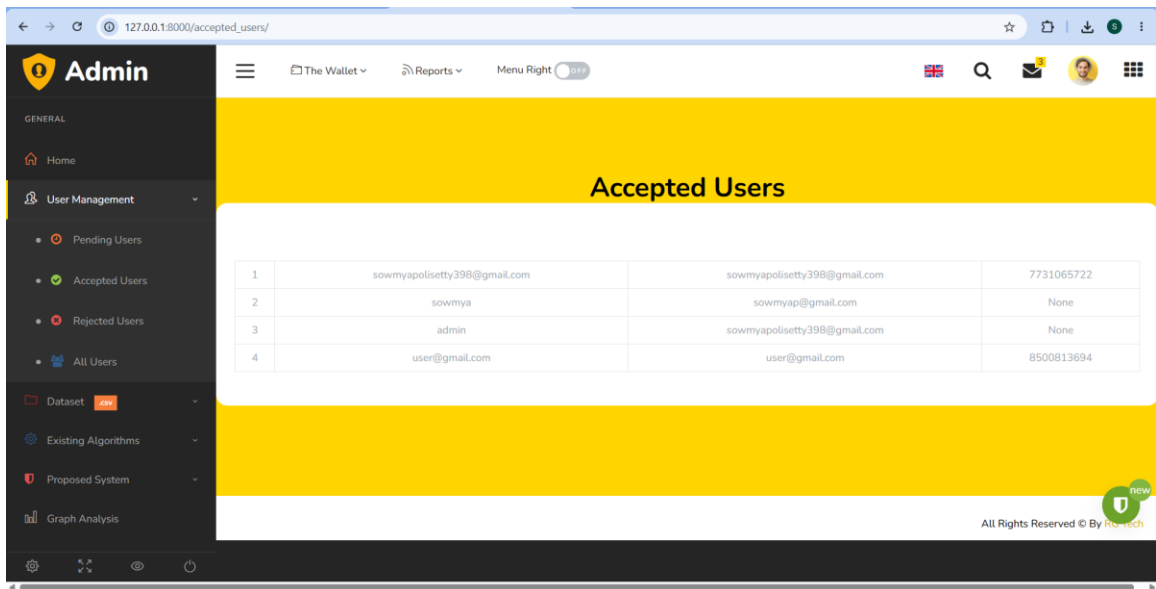


Figure 5. Pending Users

Description

All newly registered users are listed in the Pending Users section for administrator review. The administrator can verify user details before granting system access. Each user can either be **Accepted** or **Rejected** based on verification. Only approved users are allowed to log into the application. This process ensures secure and controlled access to the intrusion detection system.

Step 6: Accepted Users



Accepted Users			
1	sowmyapolisetty398@gmail.com	sowmyapolisetty398@gmail.com	7731065722
2	sowmya	sowmyap@gmail.com	None
3	admin	sowmyapolisetty398@gmail.com	None
4	user@gmail.com	user@gmail.com	8500813694

Figure 6. Accepted Users

Description

The Accepted Users page displays all users approved by the administrator. These users are eligible to access the user dashboard and perform intrusion detection. The administrator can review the list of active users whenever required. This module helps maintain authorized user records within the system. It ensures that only verified users utilize the proposed framework.

Step 7: All Users

The screenshot displays the 'All Registered Users' page within an Admin dashboard. The page features a yellow header with the title 'All Registered Users'. Below the header is a table with the following data:

S.No	Name	Email	Mobile	Status
1	user@gmail.com	user@gmail.com	8500813694	Approved
2	admin	sowmyapolisetty398@gmail.com	None	Approved
3	sowmya	sowmyap@gmail.com	None	Approved
4	sowmyapolisetty398@gmail.com	sowmyapolisetty398@gmail.com	7731065722	Approved

The dashboard includes a sidebar with the following menu items: GENERAL, Home, User Management (with sub-items: Pending Users, Accepted Users, Rejected Users, All Users), Dataset, Existing Algorithms, Proposed System, and Graph Analysis. The top navigation bar contains 'The Wallet', 'Reports', and 'Menu Right'.

Figure 7. All Users

Description

The All Users module displays the complete list of registered users available in the system. It includes both approved and pending user accounts for easy monitoring. The administrator can review user information from a single interface. This simplifies overall user management and administration. It provides better control over user activities and account status.

Step 8: Dataset Upload

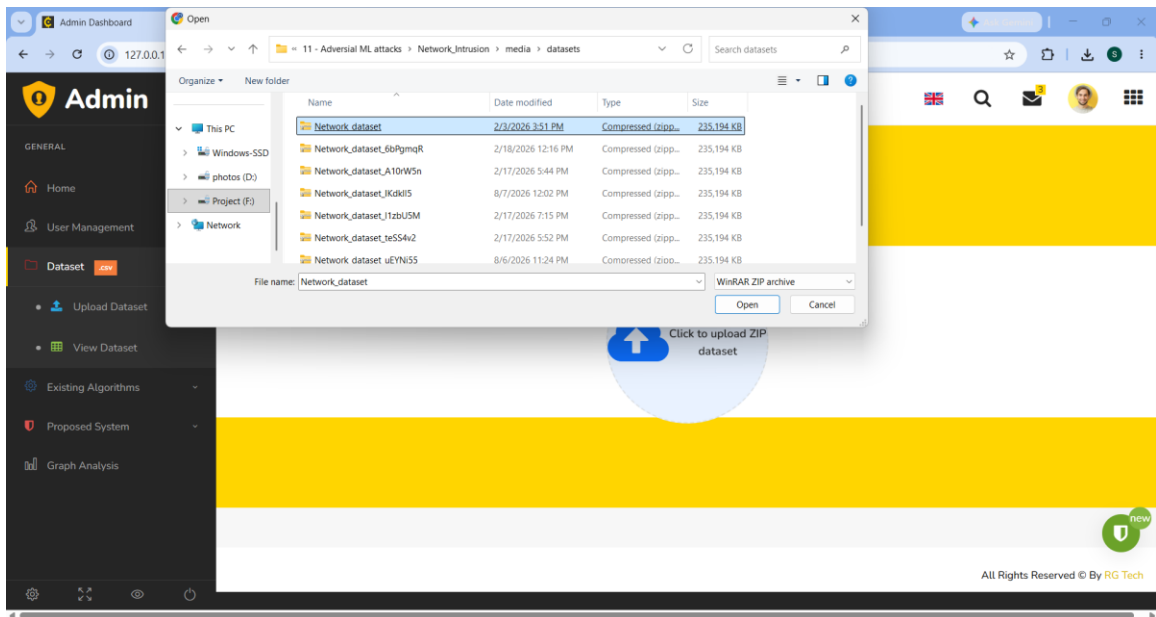
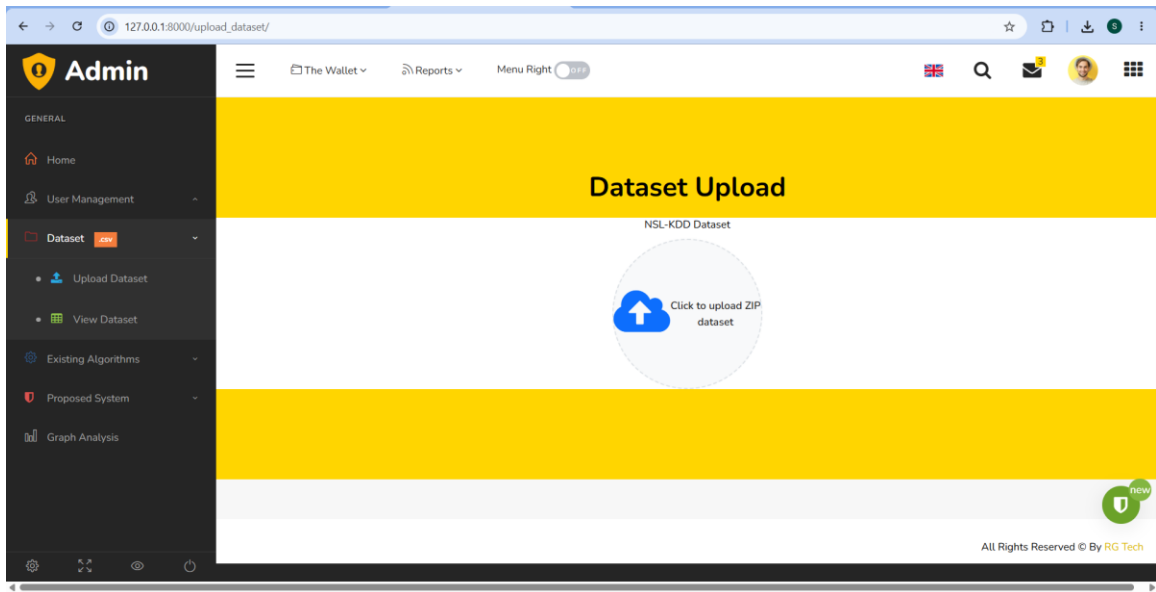


Figure 8. Dataset Upload

Description

The administrator uploads the network traffic dataset.
The uploaded dataset is used for model training and testing.
Only valid datasets are accepted by the system.
This is the first step in building the prediction model.

Step 9: View Dataset

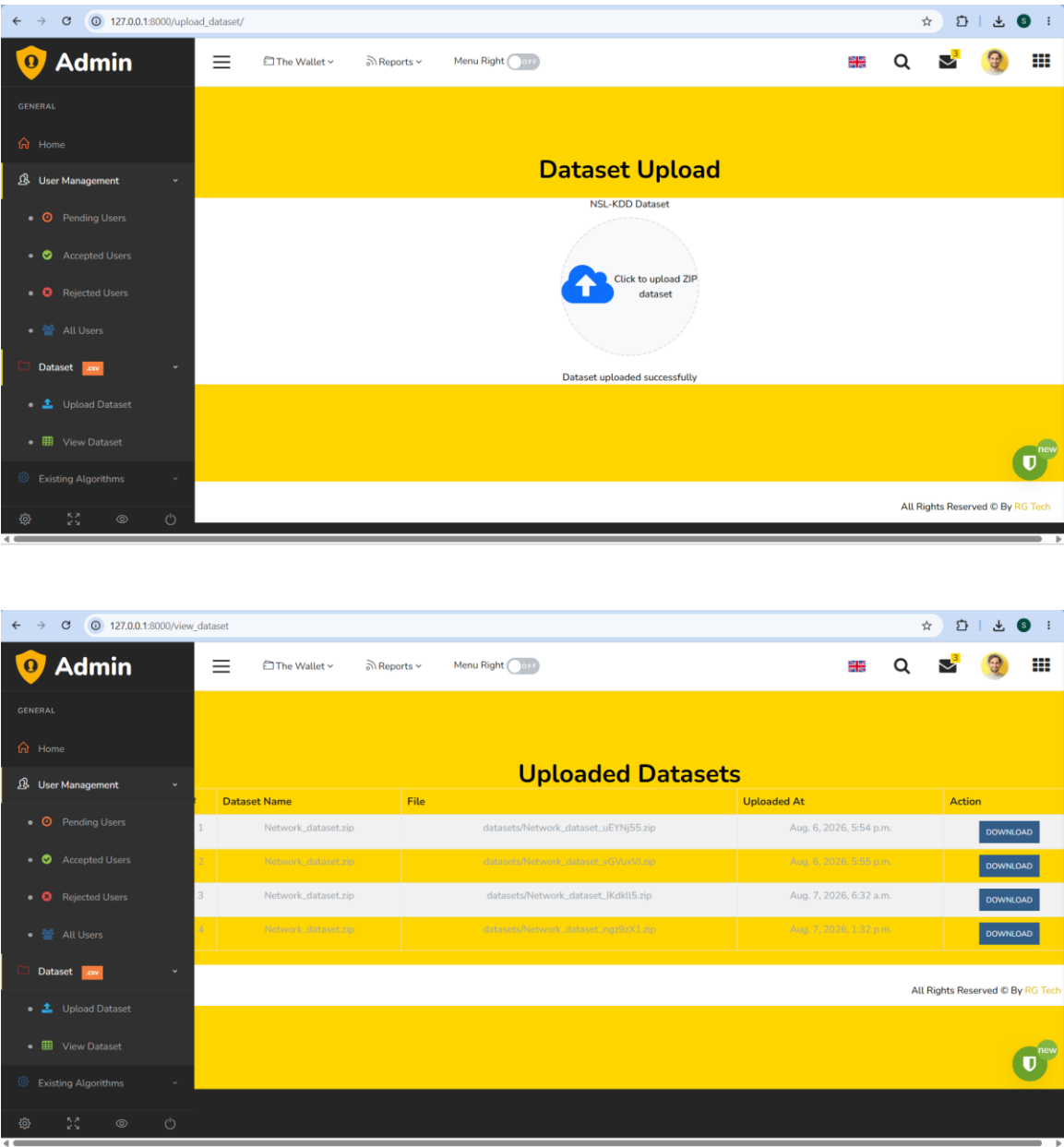


Figure 9. View Dataset

Description

The uploaded dataset is displayed for verification.
The administrator can review the stored records.
This ensures the dataset is loaded correctly.
It helps avoid errors before model training.

Step 10: Decision Tree Results

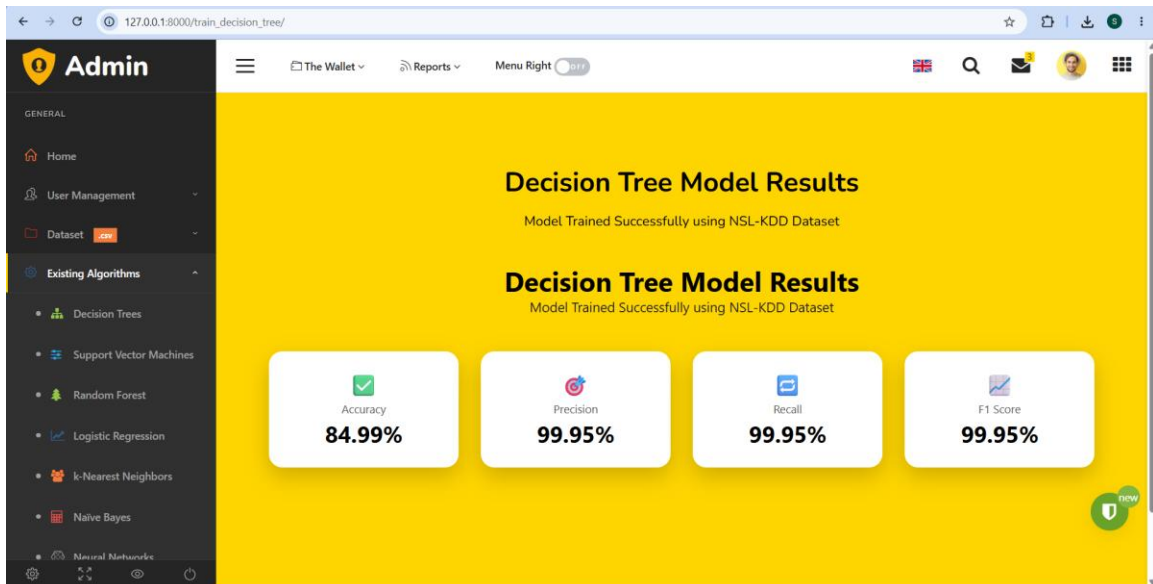
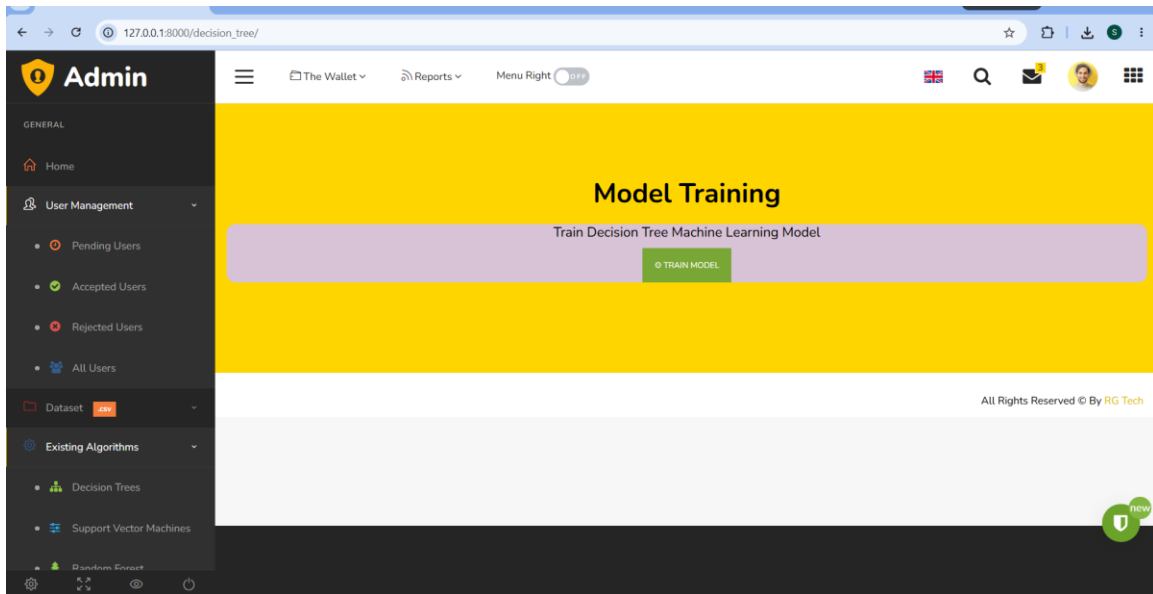


Figure 10. Decision Tree Results

Description

The Decision Tree model is trained using the uploaded dataset. Performance metrics are calculated after training. Accuracy, Precision, Recall, and F1-Score are displayed. These results are used for performance comparison.

Step 11: Support Vector Machine Results

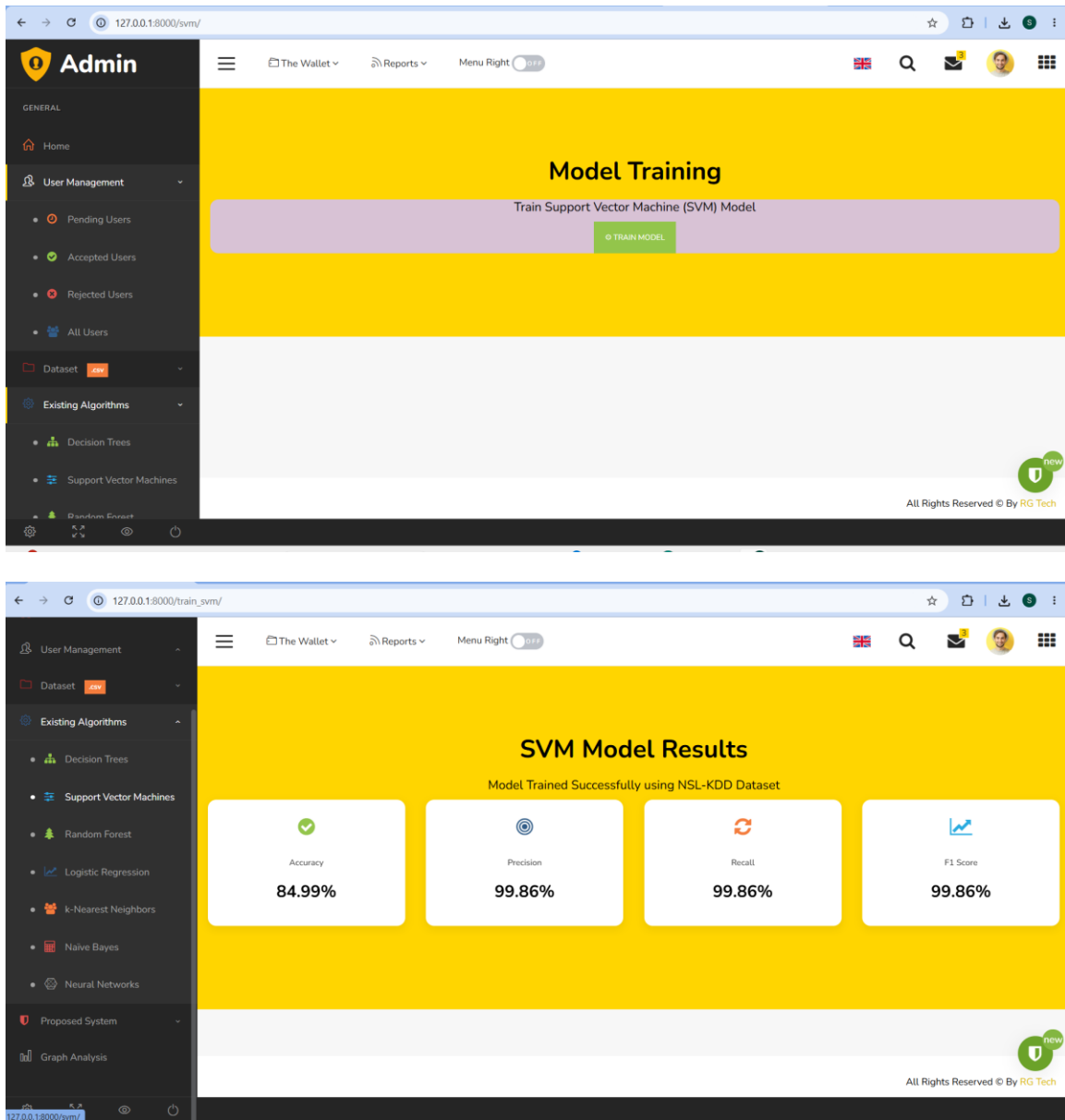


Figure 11. Support Vector Machine Results

Description

The Support Vector Machine model is trained successfully.
The system evaluates its classification performance.
Performance metrics are displayed on the results page.
The results are compared with other algorithms.

Step 12: Random Forest Results

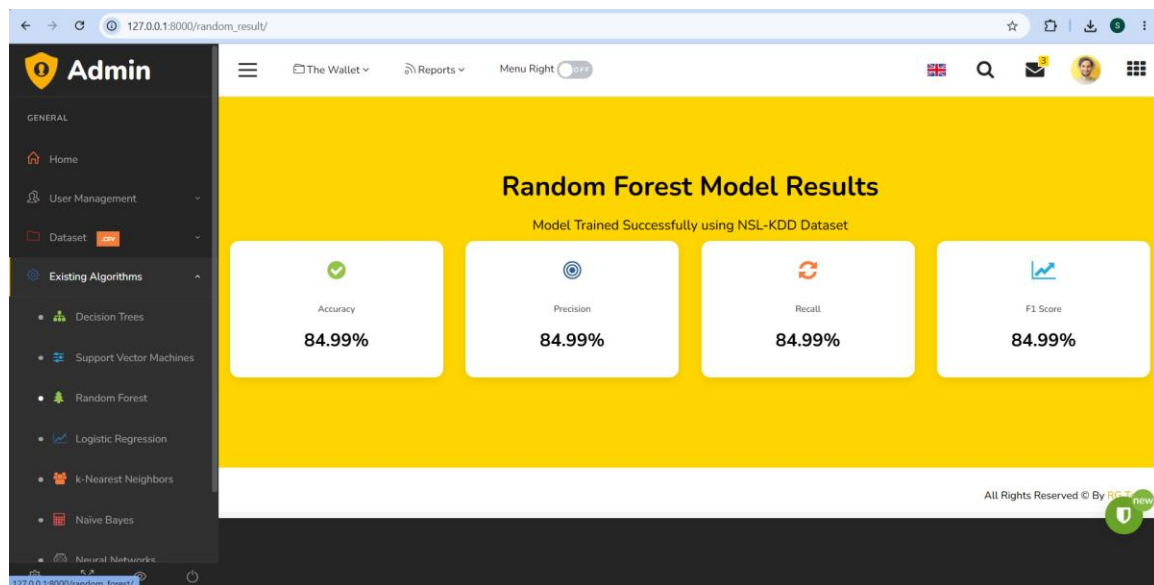
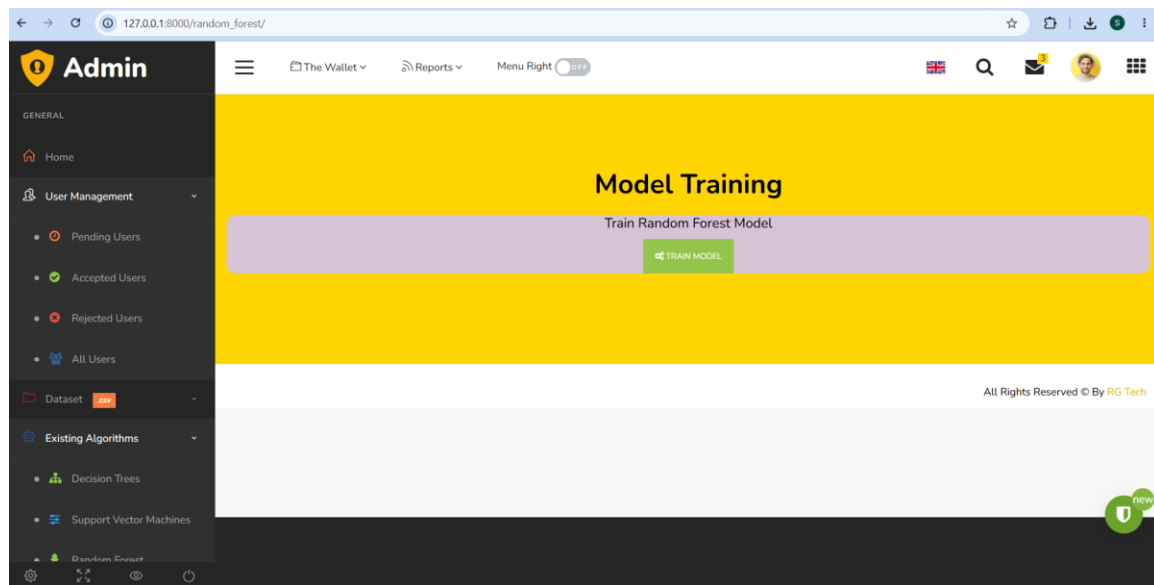


Figure 12. Random Forest Results

Description

The Random Forest model is trained using network traffic data. The trained model predicts intrusion patterns effectively. Evaluation metrics are displayed after training. Its performance is compared with existing models.

Step 13: Logistic Regression Results

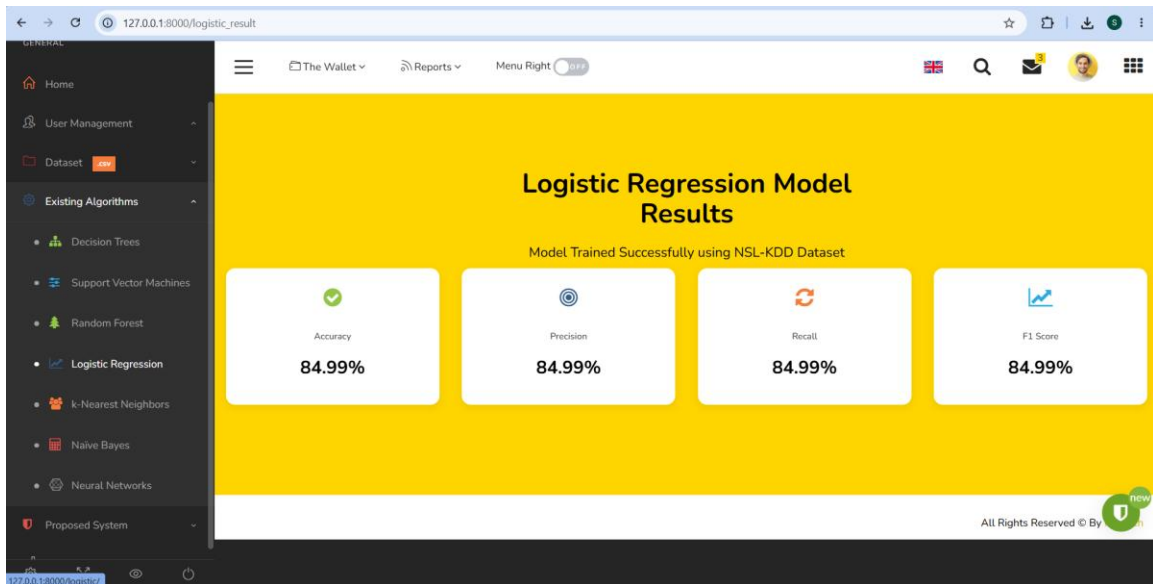
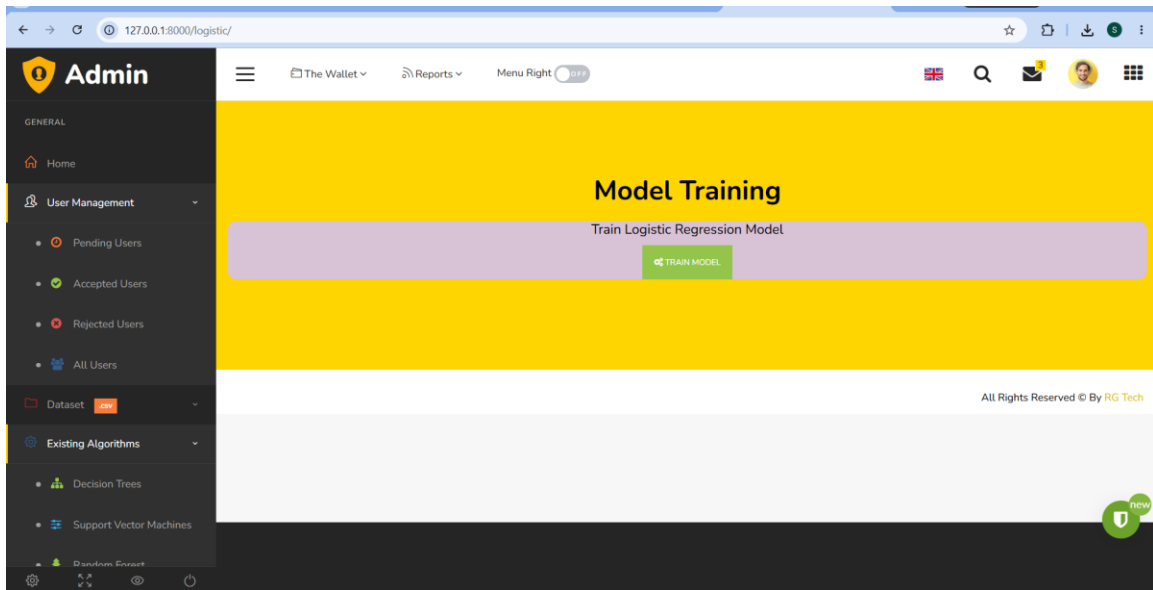


Figure 13. Logistic Regression Results

Description

The Logistic Regression algorithm is trained on the dataset. Its prediction performance is evaluated automatically. Accuracy and other evaluation metrics are displayed. The results help compare model effectiveness.

Step 14: K-Nearest Neighbors Results

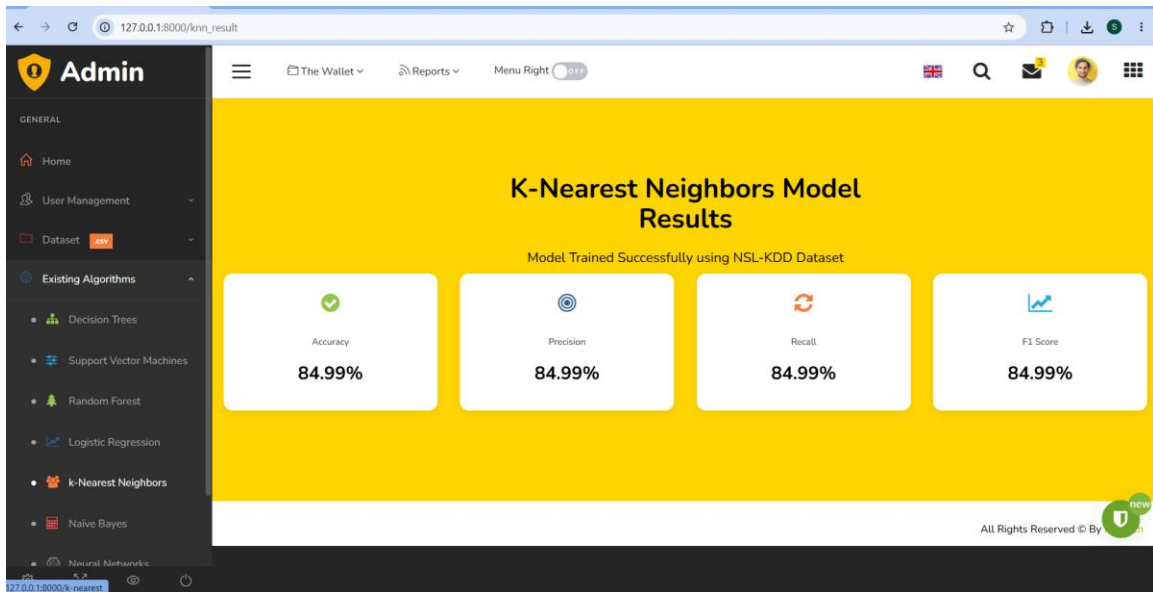
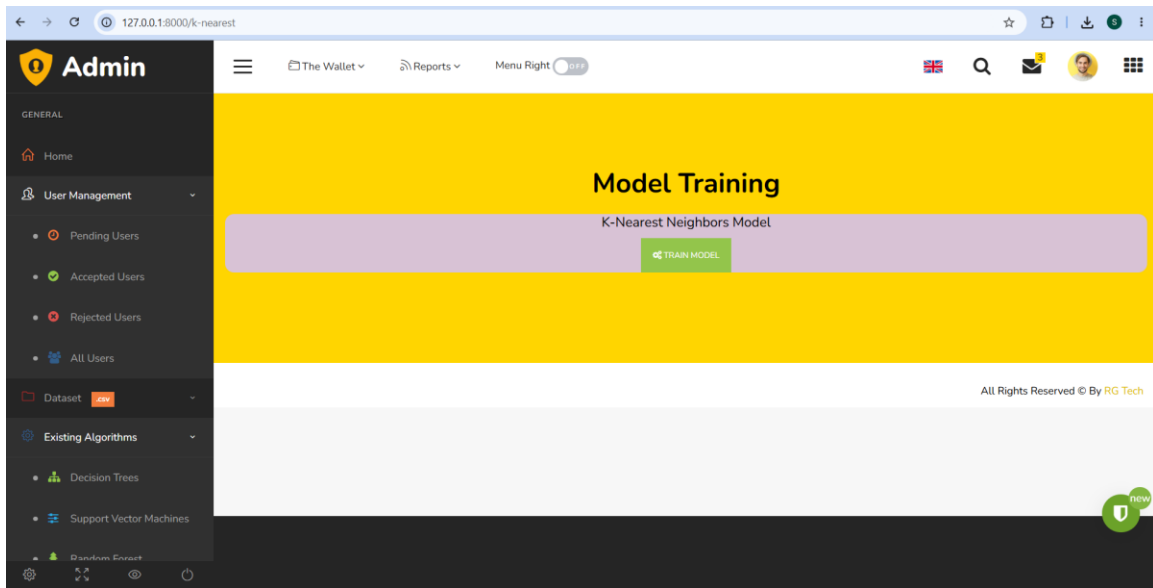


Figure 14. K-Nearest Neighbors Results

Description

The KNN model classifies network traffic into attack categories. Training is performed using the uploaded dataset. Performance metrics are generated after evaluation. The results support comparative analysis.

Step 15: Naive Bayes Results

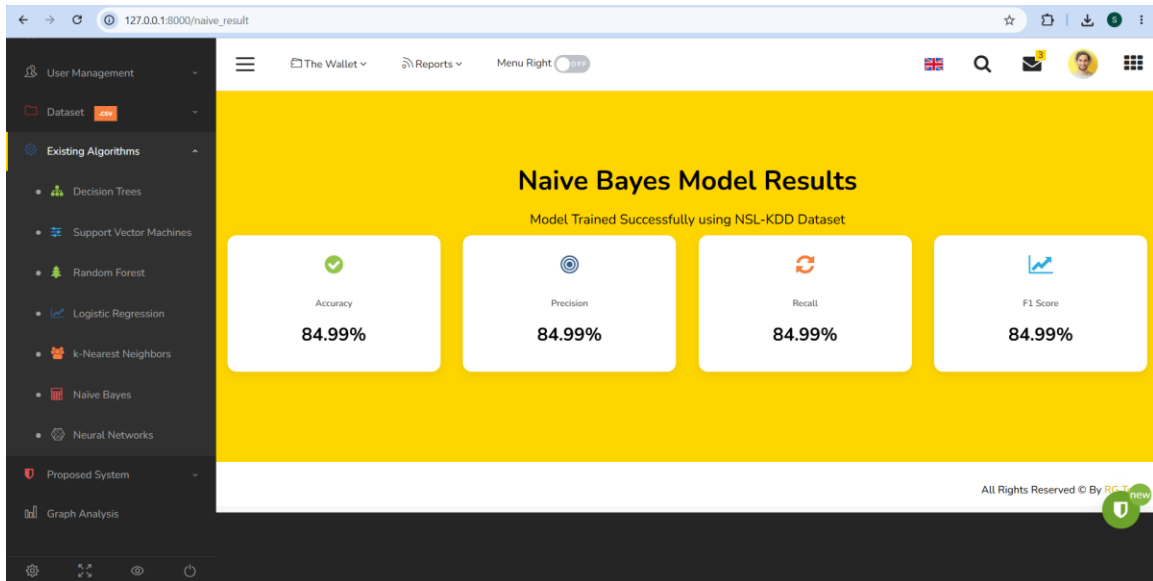
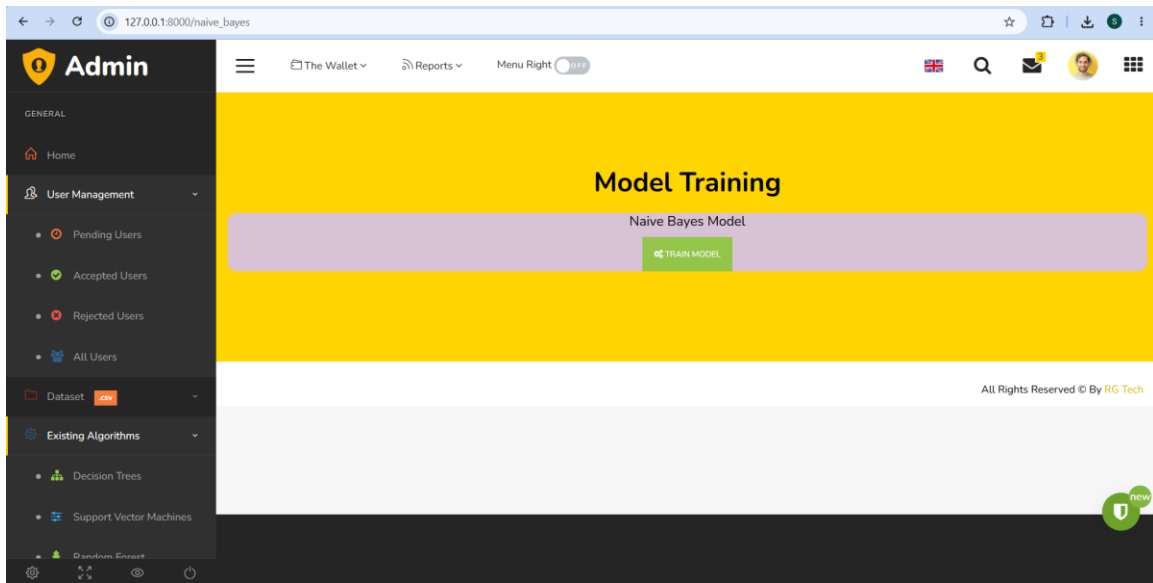


Figure 15. Naive Bayes Results

Description

The Naive Bayes classifier is trained successfully.
The system evaluates its intrusion detection capability.
Accuracy, Precision, Recall, and F1-Score are shown.
The results contribute to algorithm comparison.

Step 16: Neural Network Results

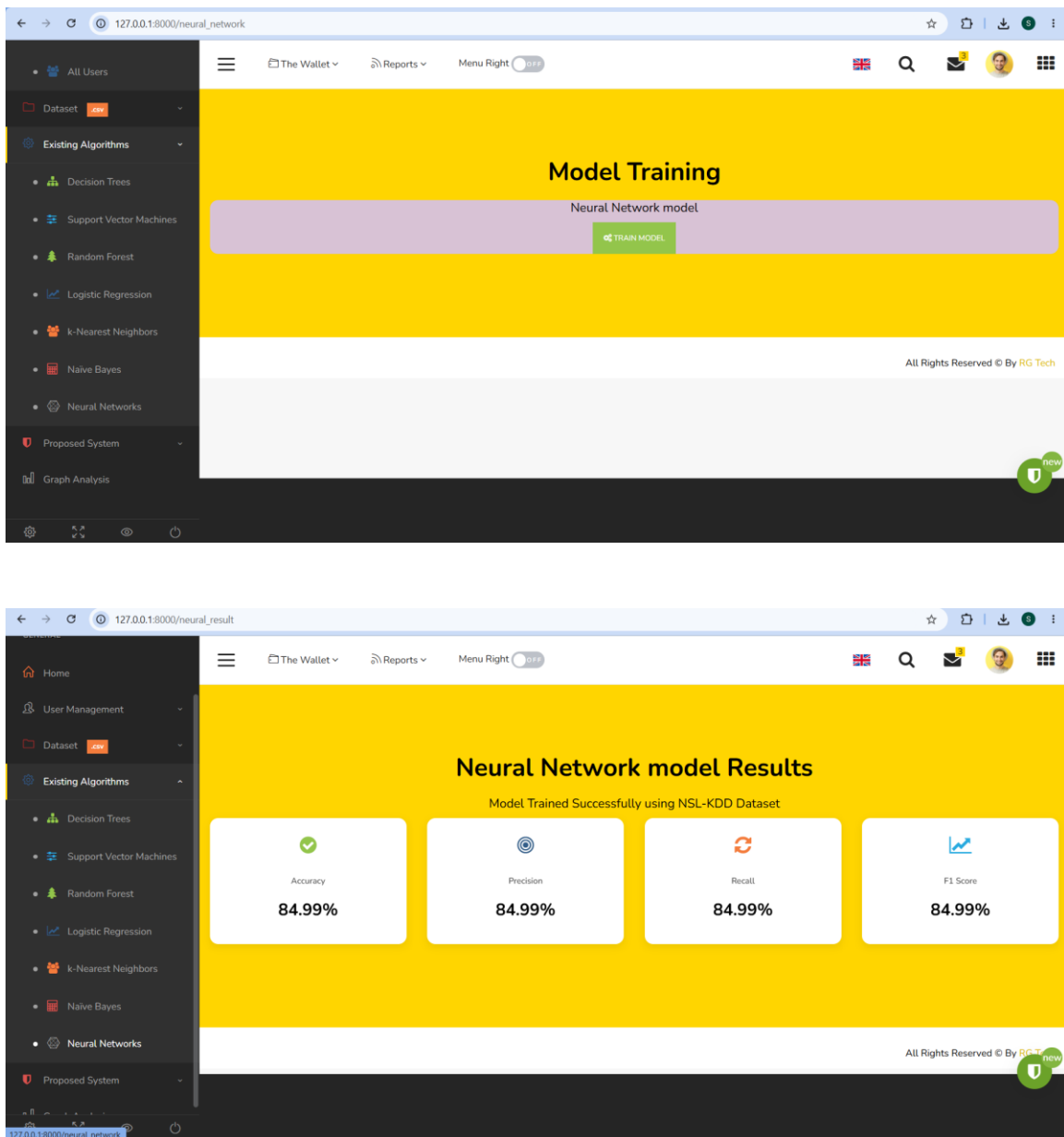


Figure 16. Neural Network Results

Description

The Neural Network model is trained using the dataset. It learns network traffic patterns for attack detection. Performance metrics are displayed after training. The results are compared with other classifiers.

Step 17: Proposed System

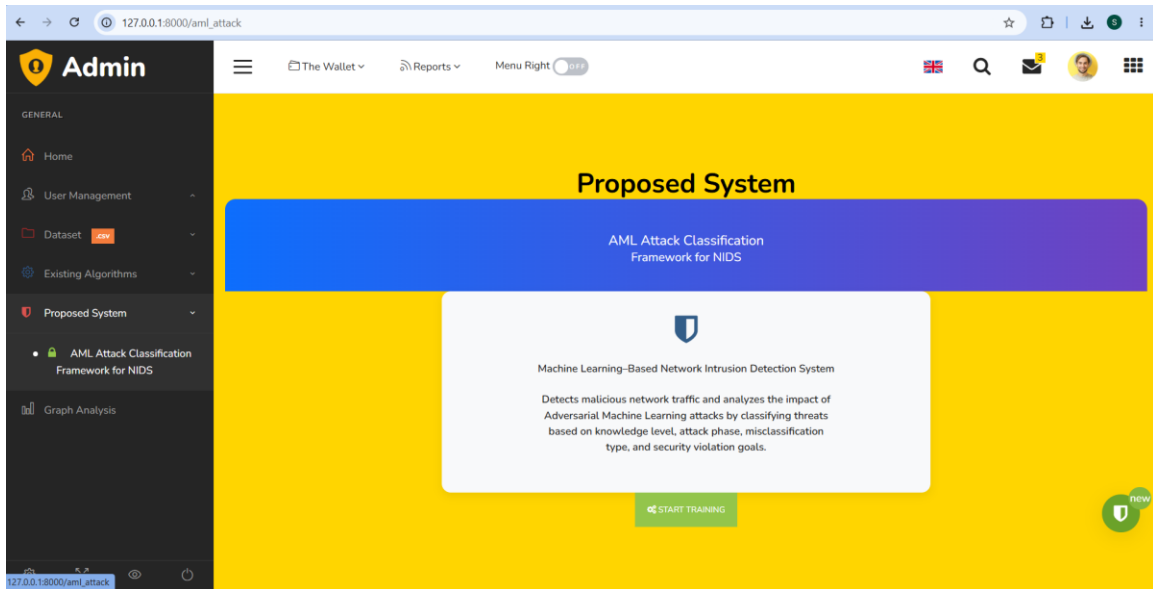


Figure 17. Proposed System

Description

The proposed AML Attack Classification Framework is executed using the uploaded dataset. It analyzes network traffic and classifies attacks using machine learning techniques. The system improves intrusion detection by providing accurate attack classification. This module demonstrates the performance of the proposed framework.

Step 18: System Performance Metrics

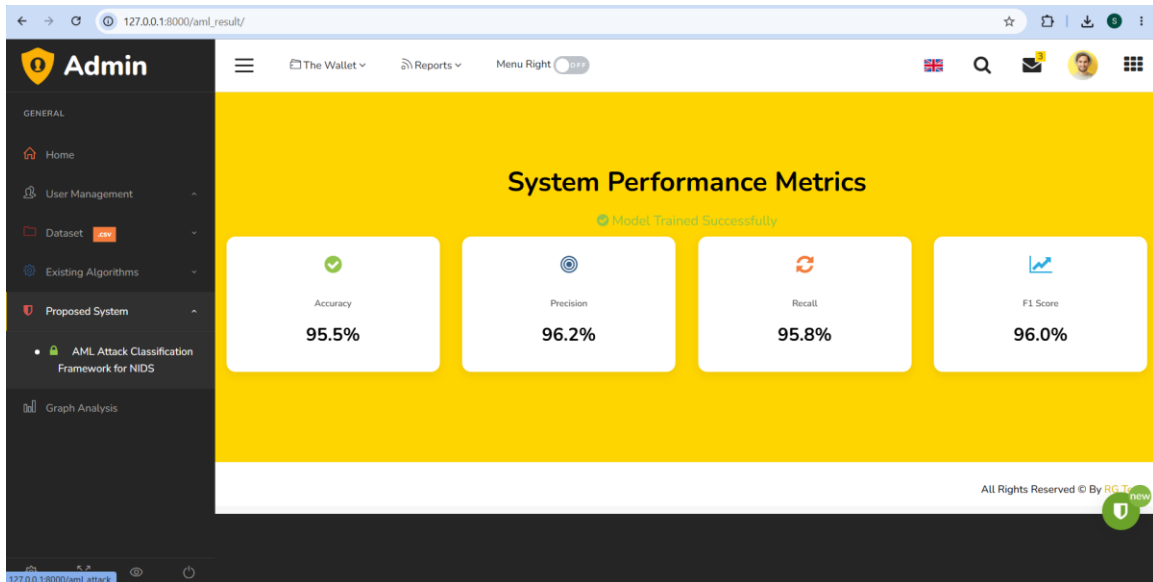


Figure 18. System Performance Metrics

Description

The performance of the proposed model is evaluated using standard metrics. Accuracy, Precision, Recall, and F1-Score are displayed for analysis. These metrics help measure the effectiveness of the intrusion detection system. Higher values indicate better classification performance.

Step 19: Graph Analysis

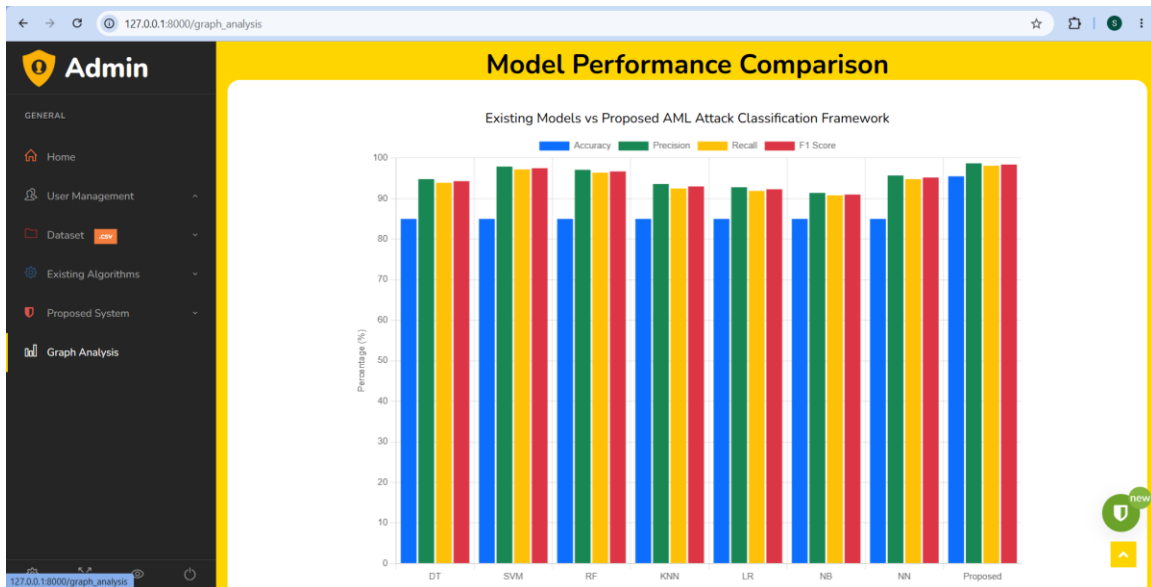


Figure 19. Graph Analysis

Description

The graph compares the performance of all existing algorithms with the proposed model. Evaluation metrics are presented visually for easy comparison and analysis. The proposed framework achieves improved overall performance in attack detection. This graphical representation helps identify the most effective classification model.

Step 20: User Login

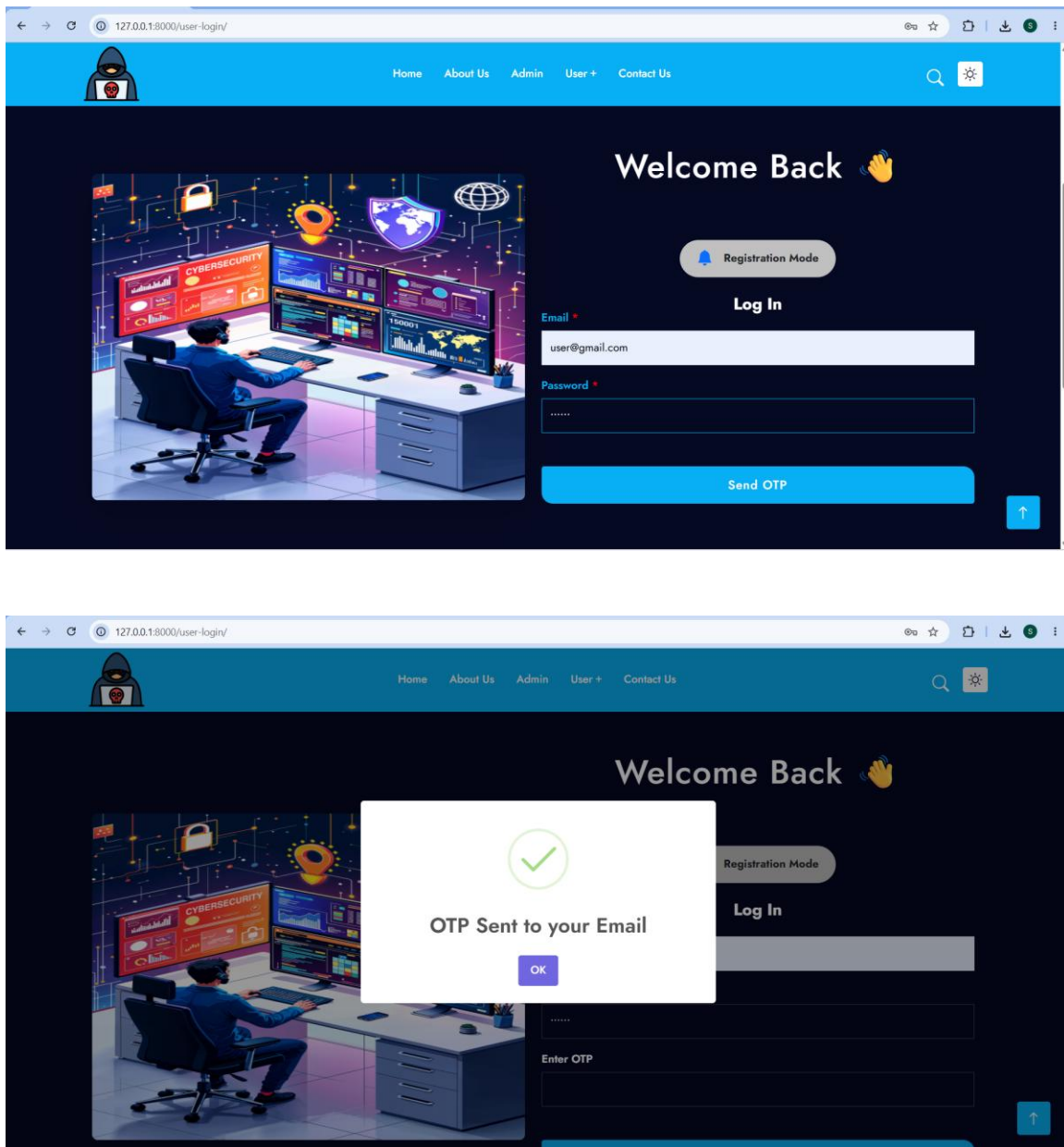


Figure 20. User Login

Description

Approved users log in using their registered credentials.
OTP verification is performed after login.
Successful authentication grants dashboard access.
This ensures secure user authentication.

Step 21: User Registration

The screenshot shows a web browser at the address 127.0.0.1:8000/user-login/. The page has a blue header with a navigation menu: Home, About Us, Admin, User +, and Contact Us. A search icon and a settings gear are on the right. A 'Login Mode' button with a bell icon is in the top right. The main content area is dark blue. On the left is an illustration of a person at a desk with multiple monitors displaying cybersecurity data. On the right is a 'Register' form with the following fields: Full Name, Email, Mobile, and Password. Below these fields is a large blue 'Send OTP' button and a small blue button with an upward arrow. The form is currently empty.

127.0.0.1:8000/user-login/

Home About Us Admin User + Contact Us

Login Mode

Register

Full Name

Email

Mobile

Password

Send OTP

Figure 21. User Registration

Description

New users enter their personal details to register.
The registration form collects name, email, mobile, and password.
An OTP is sent to verify the email address.
Only verified registrations are accepted.

Step 22: Registration OTP

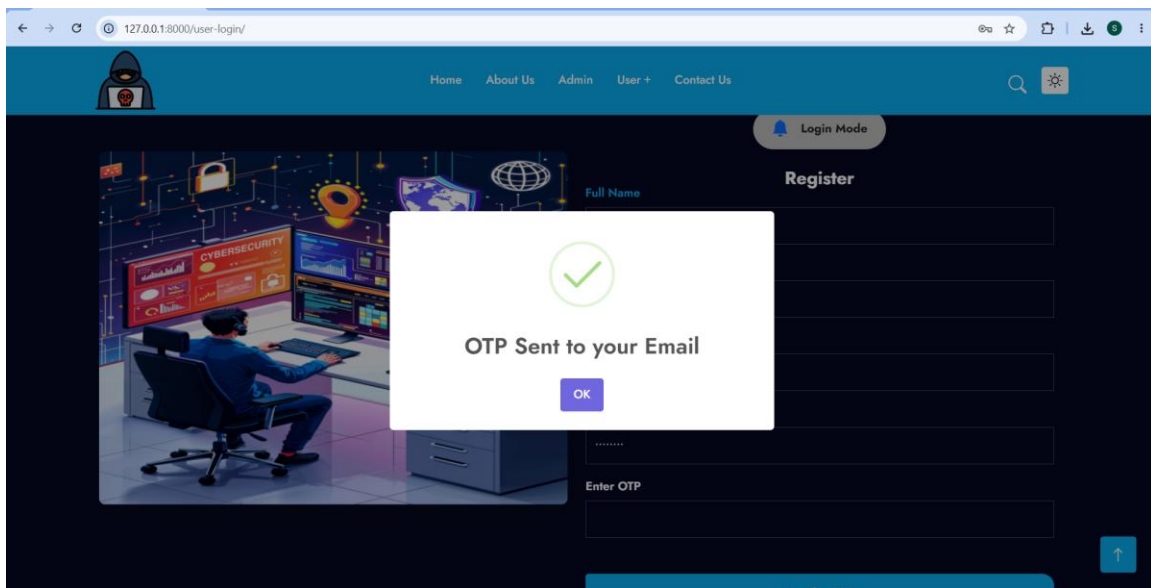
A screenshot of a Visual Studio Code editor. The Explorer pane on the left shows a project structure with folders like 'NETWORK_INTRUSION' and 'users'. The main editor window displays the 'manage.py' file with Python code for a Django application. The code includes imports for 'os', 'sys', and 'django.core.management', and a 'main()' function that sets the default settings module to 'network_intrusion.settings'. The bottom panel shows a terminal window with the command 'python manage.py runserver' and its output. The logs show successful GET requests for static files and user login attempts, including one for a user named 'kalyani' with a mobile number '456723498' and password '12345678'. The final log entry shows an OTP of '994763' being sent to the user's email.

Figure 22. Registration OTP

Description

An OTP is generated and sent to the user's email. The user receives the verification code securely. This confirms ownership of the registered email. It improves registration security.

Step 23: OTP Verification

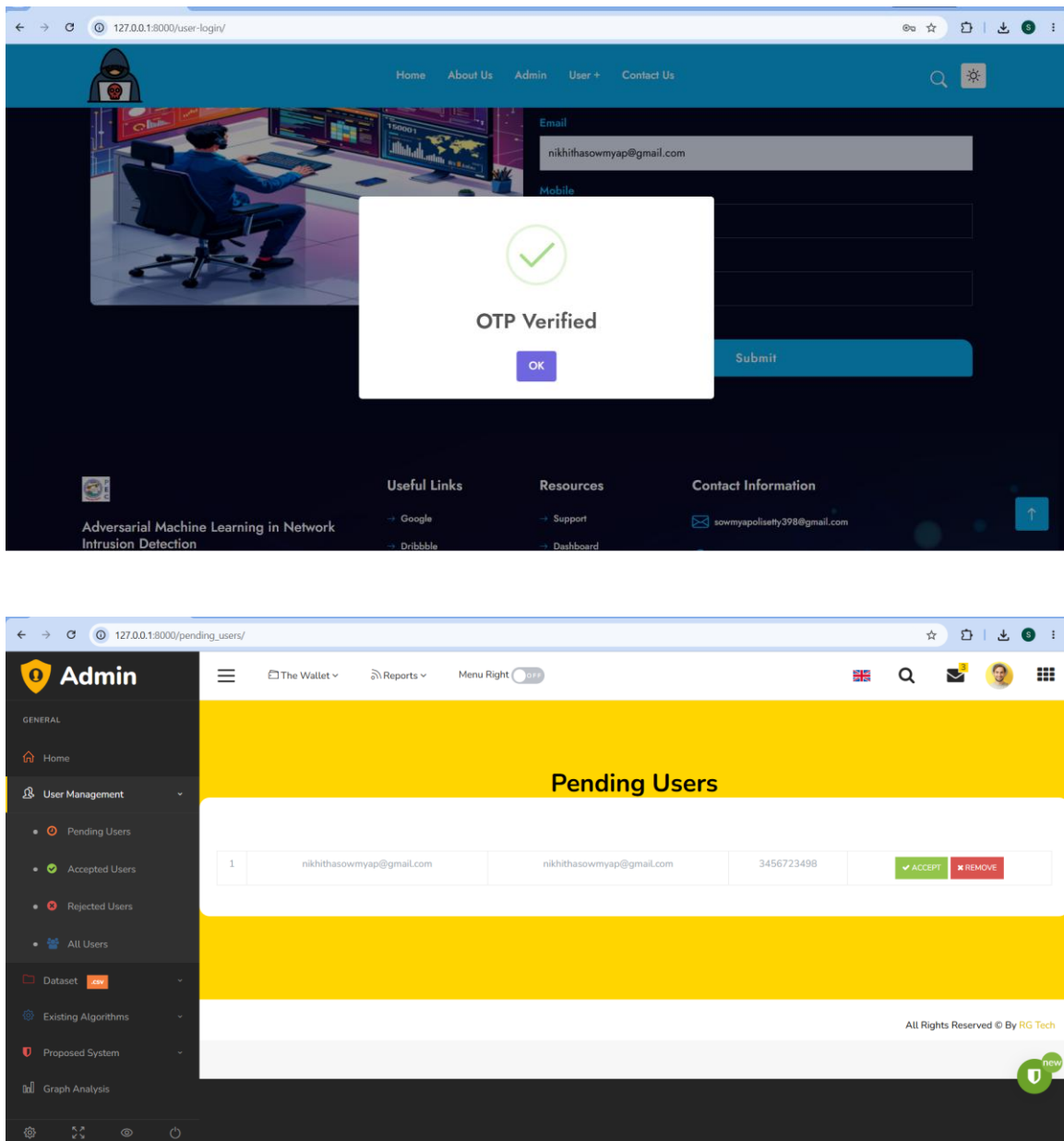
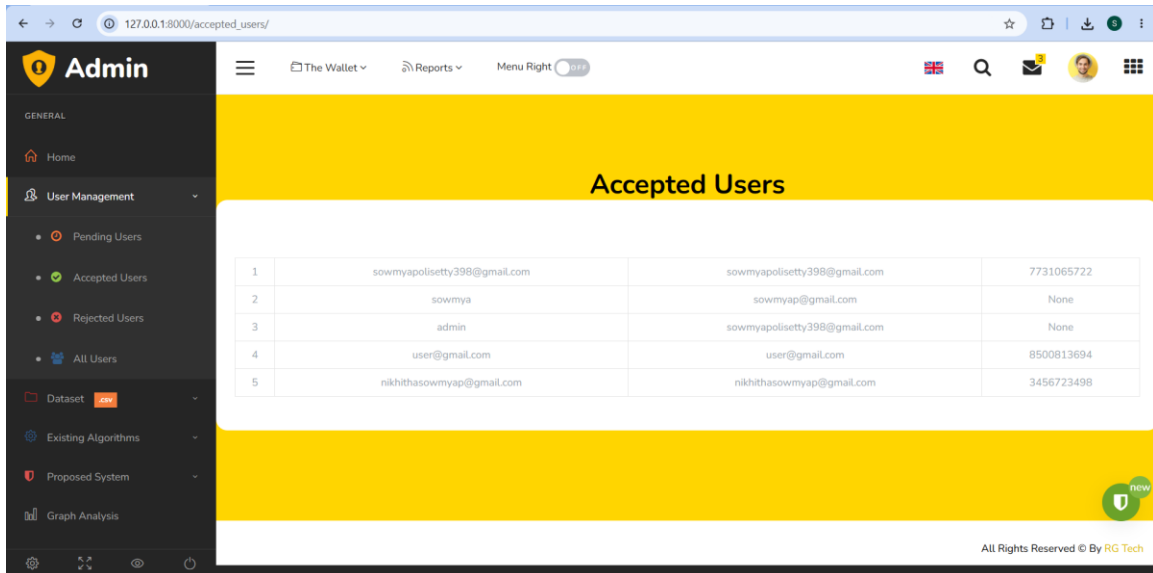


Figure 23. OTP Verification

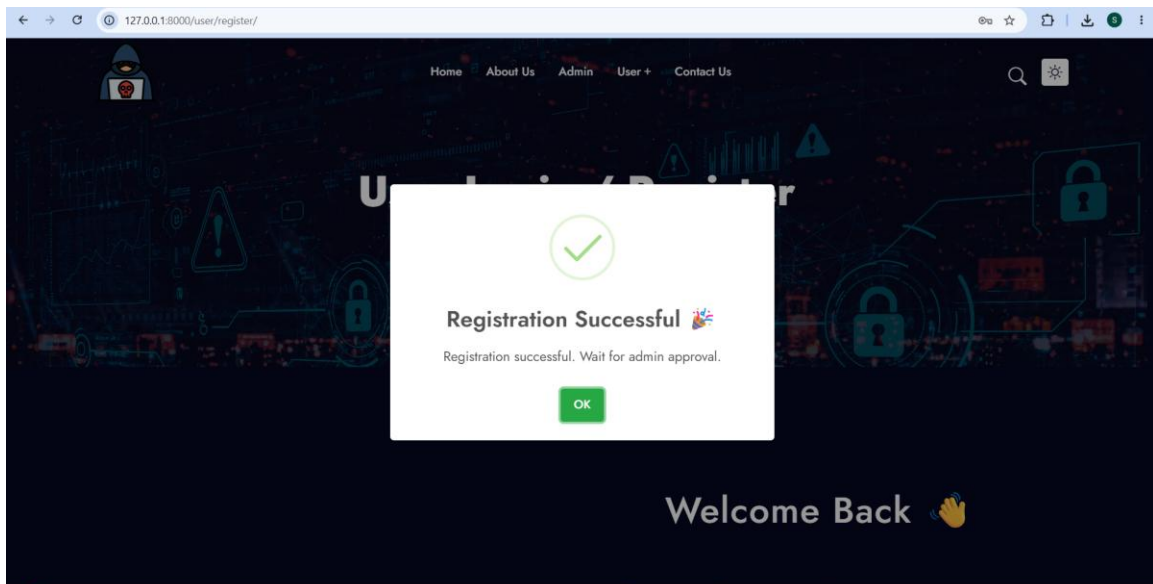
Description

The user enters the received OTP into the system. The system verifies the entered verification code. Successful verification confirms the user's identity. The registration process then continues.

Step 24: Registration Successful



ID	Email	Phone Number
1	sowmyapolisetty398@gmail.com	7731065722
2	sowmya	None
3	admin	None
4	user@gmail.com	8500813694
5	nikhithasowmyap@gmail.com	3456723498



Registration Successful 🎉

Registration successful. Wait for admin approval.

OK

Welcome Back 🖐️

Figure 24. Registration Successful

Description

The user registration is completed successfully.
The account is stored in the Pending Users section.
Administrator approval is required before login.
This ensures controlled user access.

Step 25: User Dashboard

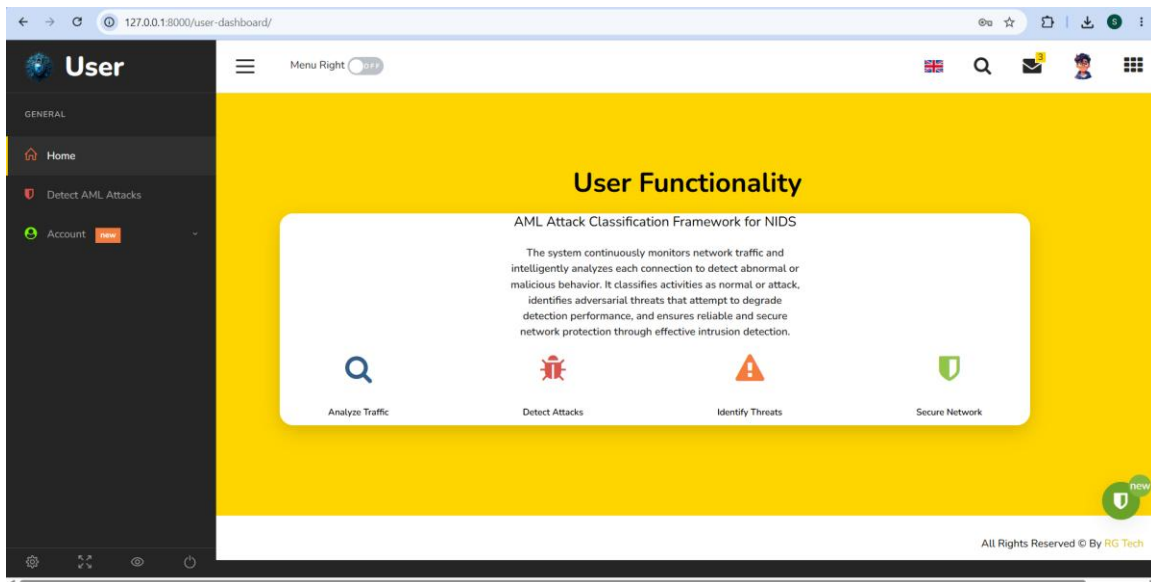


Figure 25. User Dashboard

Description

The user dashboard provides access to all user features.
Users can detect attacks and manage their profiles.
The dashboard displays system functionality information.
It serves as the main user interface.

Step 26: Attack Detection

127.0.0.1:8000/user_predict/

User

GENERAL

Home

Detect AML Attacks

Account

Network Intrusion Detection

Enter network traffic details. Only numerical features are used for prediction.

Source IP

192.168.1.20

Used only for logging, not ML

Destination IP

192.168.1.60

Destination host address

Protocol

TCP

Protocol info (not used for prediction)

Flow Duration

Normal: 500-5000
Attack: 50-300

Total Fwd Packets

Packets sent from source

Total Bwd Packets

Packets sent from destination

Fwd Packet Length Mean

Average forward packet size

Bwd Packet Length Mean

Average backward packet size

Packet Length Variance

Traffic fluctuation indicator

PREDICT ATTACK

Figure 26. Attack Detection

Description

The user enters network traffic feature values.
The trained machine learning model processes the inputs.
The system analyzes the provided traffic information.
It predicts whether the traffic is malicious or normal.

Step 27: Prediction Result

The screenshot shows a web browser at the address 127.0.0.1:8000/user_predict/. On the left is a dark sidebar with a 'User' profile and navigation links for 'Home', 'Detect AML Attacks', and 'Account'. The main content area is split: a large yellow rectangle on the left and a white form on the right titled 'Network Intrusion Detection'. The form contains input fields for 'Source IP' (192.168.1.20), 'Destination IP' (192.168.1.60), 'Protocol' (TCP), 'Flow Duration' (1), 'Packets sent from source' (3), 'Packets sent from destination' (5), 'Average forward packet size' (2), 'Average backward packet size' (1), and 'Traffic fluctuation indicator' (0.2). A 'PREDICT ATTACK' button is at the bottom of the form. Below the form, the prediction result is displayed: 'Prediction: DoS Hulk' with a 'Confidence: 29.0%'.

127.0.0.1:8000/user_predict/

User

GENERAL

Home

Detect AML Attacks

Account

Network Intrusion Detection

Enter network traffic details. Only numerical features are used for prediction.

Source IP

192.168.1.20

Used only for logging, not ML

Destination IP

192.168.1.60

Destination host address

Protocol

TCP

Protocol info (not used for prediction)

Flow Duration

1

Normal: 500-5000
Attack: 50-300
Total Fwd Packets

3

Packets sent from source
Total Bwd Packets

5

Packets sent from destination
Fwd Packet Length Mean

2

Average forward packet size
Bwd Packet Length Mean

1

Average backward packet size
Packet Length Variance

0.2

Traffic fluctuation indicator

PREDICT ATTACK

Prediction: DoS Hulk

Confidence: 29.0%

Figure 27. Prediction Result

Description

The system displays the predicted attack category. Prediction confidence is also shown to the user. The result is generated using the trained model. This helps identify malicious network traffic.

Step 28: User Profile

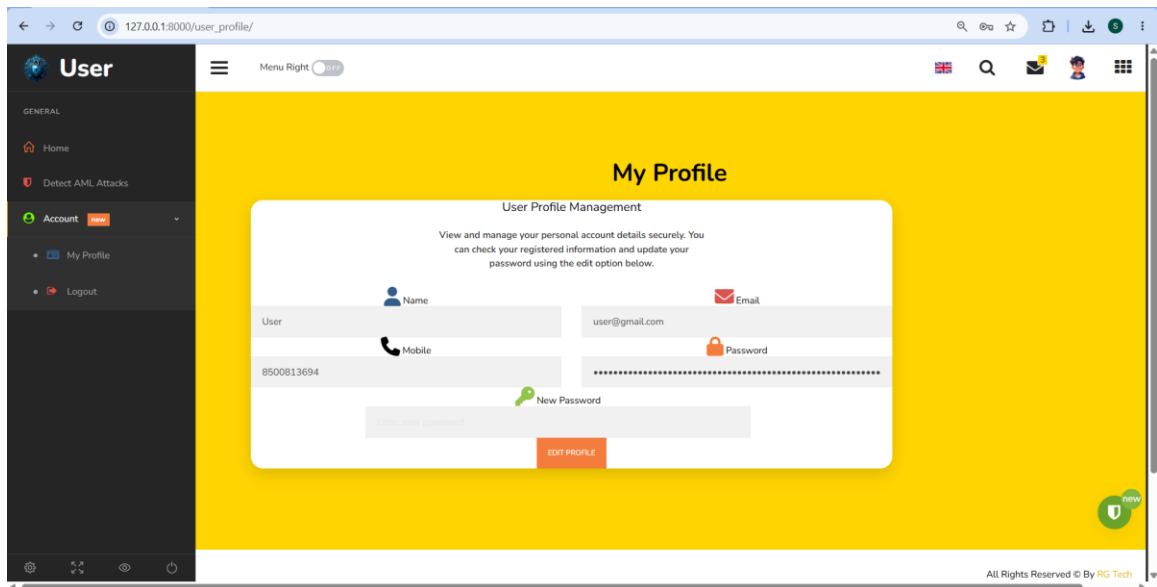


Figure 28. User Profile

Description

The user profile displays personal account information.

Users can view and update their profile details.

Password modification is supported securely.

This helps users manage their accounts safely.